# Matthew Kalarickal

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## Education

**Worcester Polytechnic Institute** — Worcester, MA
*Master of Science in Computer Science (In Progress), GPA: 4.0/4.0* — *Aug. 2025 – Present*

**College of the Holy Cross** — Worcester, MA
*B.A. in Computer Science and Mathematics, magna cum laude, GPA: 3.83/4.00* — *Sep. 2021 – May 2025*

## Experience

**AI Intern** — June 2024 – Aug. 2024
*Camgian* — *Tuscaloosa, AL*

- Designed and trained supervised neural network models for real-time decision making in adversarial simulation environments
- Engineered feature extraction pipelines processing large-scale simulation logs to model engagement behavior
- Developed adversarial simulation scenarios in AFSIM to evaluate and improve autonomous decision-making performance
- Integrated predictive models into mission-planning simulation pipelines, ensuring robustness and usability

**Teaching Assistant** — Sep. 2022 – Present
*Worcester Polytechnic Institute; College of the Holy Cross* — *Worcester, MA*

- Supported 100+ students across computer science and mathematics courses through office hours, labs, and individualized technical guidance
- Guided students in algorithm design, debugging, software engineering practices, and mathematical problem solving
- Evaluated programming assignments for correctness, efficiency, and design quality

## Research Experience

**Evaluating Pedagogical Styles of LLM-Generated Hint Sets** — Jan. 2026 – Present

- Designed prompting frameworks to generate mathematics hint sets following multiple instructional strategies
- Developed an evaluation framework comparing pedagogical styles of LLM-generated mathematics hints

**Fairness of LLM Feedback** — Sep. 2025 – Present

- Developed a benchmarking pipeline evaluating fairness of LLM-generated educational feedback across English dialects
- Applied transformer embeddings and statistical hypothesis testing to identify systematic dialect bias across multiple LLMs
- Designed controlled experiments isolating prompt wording and dialect effects on model-generated feedback

**Honors Thesis — Multi-Agent Reinforcement Learning** — Sep. 2024 – May 2025

- Designed communication protocols enabling coordination among cooperative reinforcement learning agents
- Implemented multi-agent learning algorithms under constrained communication settings
- Evaluated emergent communication behavior and coordination performance across training regimes

## Publications

**Matthew Kalarickal**, et al., Evaluating Pedagogical Styles of LLM-Generated Hint Sets. *Artificial Intelligence in Education (AIED) 2026 Doctoral Consortium*, pp. 390-395.

## Technical Skills

**Languages**: Python, C++, Java, SQL, R
**Machine Learning**: PyTorch, TensorFlow, scikit-learn (supervised/unsupervised learning)
**GenAI & LLM Systems**: LLM evaluation & benchmarking, prompt engineering, transformer embeddings, Hugging Face Transformers, statistical validation
**Data & Tools**: pandas, NumPy, Matplotlib, Git, VS Code, Visual Studio, PyCharm

## Awards & Honors

Pi Mu Epsilon (National Mathematics Honor Society) | Putnam Competition (Score: 10) | Uechi-Ryu Karate, Third Degree Black Belt